repeat this argument and after finitely many steps we cover all of M

so $\|f\|_{L^1(M)} \leq C$ $\square$

<u>Step 2</u> Bounding $\|f\|_{C^0}$, i.e. $\inf_M f$ from below

<u>Preliminary theorem</u> (Alexandrov-Bakelman-Pucci)

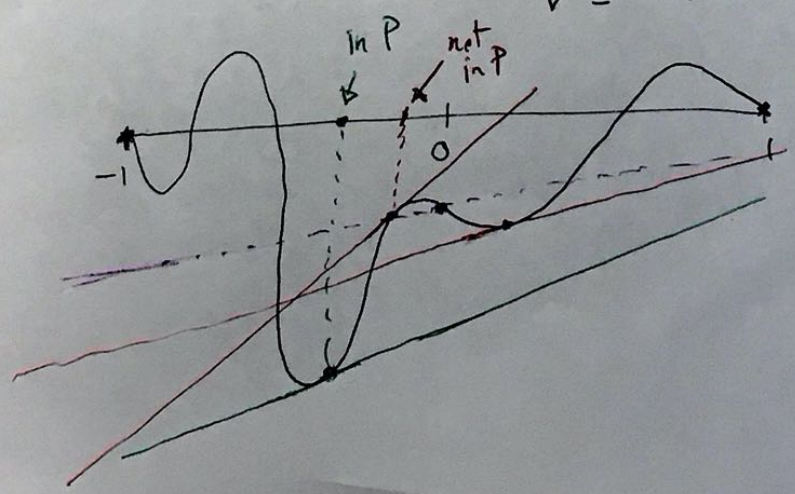
Let $v : B_1 \longrightarrow \mathbb{R}$ be smooth ($B_1 \subset \mathbb{R}^n$ unit ball)

s.t. $v(0) + \varepsilon \leq \inf_{\partial B_1} v$ for some $\varepsilon > 0$

Let $P = \left\{ x \in B_1 \;\middle|\; \begin{array}{l} v(y) \geq v(x) + Dv(x)\cdot(y-x) \\ \text{for all } y \in B_1, \\ |Dv(x)| \leq \varepsilon/2 \end{array} \right\}$

Then $\int_P \det(D^2 v) \geq c_n \varepsilon^n$ for a constant $c_n > 0$ dep. only on n

<u>Note</u>: on P, $D^2 v \geq 0$
$|Dv| \leq \varepsilon/2$
$v \leq v(0) + \varepsilon/2$



<u>Proof</u> Let $\xi \in \mathbb{R}^n$, $|\xi| \leq \varepsilon/2$

$\ell(x) = v(0) + \xi . x$

since $v(0) \leq \inf_{\partial B_1} v - \varepsilon$

graph of ℓ lies below graph of v on ∂B_1

$\Rightarrow \exists\, k \geq 0$ s.t graph of $\ell(x) - k$ is tangent to graph of v at some $x \in B_1$ and lies below the graph of v

Then $x \in P$, $Dv(x) = \xi$ and $D^2 v(x) \geq 0$

So $B(0, \varepsilon/2) \subset Dv(P)$

$(Dv : B_1 \to \mathbb{R}^n)$

$$c_n \varepsilon^n \leq \mathrm{Vol}(Dv(P)) = \int_P \det(D^2 v) \qquad \square$$

<u>Bounding $\inf_M f$</u>: we work in local coordinates s.t. $f(0) = \inf_M f$

Let $v = f + \varepsilon |z|^2$ ($\varepsilon > 0$ to be chosen)

ABP $\Rightarrow \int_P \det(D^2 v) \geq c_{2n} \varepsilon^{2n}$

on P: $D^2 v \geq 0$

$$\Longrightarrow \det(D^2 v) \leq 2^{2n} \det(\partial_j \partial_{\bar{k}} v)^2$$

$$f_{j\bar{k}} \geq -\varepsilon \delta_{j\bar{k}}$$

If $\varepsilon > 0$ is small enough (dep. on (M, J))

then $g_{j\bar{k}} + \partial_j \partial_{\bar{k}} f \geq \frac{1}{2} g_{j\bar{k}}$ on P

$$\Longrightarrow \quad \partial_j \partial_{\bar{k}} v \leq \partial_j \partial_{\bar{k}} f + C_1 g_{j\bar{k}}$$
$$\leq C_2 (g_{j\bar{k}} + \partial_j \partial_{\bar{k}} f)$$

$$\Rightarrow \det(\partial_j \partial_{\bar{k}} v) \leq C_3 \underbrace{\det(g_{j\bar{k}} + \partial_j \partial_{\bar{k}} f)}_{e^F \det(g_{j\bar{k}})} \leq C_4 \quad (\text{on } P)$$

$$\mathrm{Vol}(P) \geq C_5^{-1}$$

On P we have

$$f \leq v \leq v(0) + \varepsilon/2 \leq \inf_M f + \varepsilon/2$$

$$\Rightarrow \|f\|_{L^1} \geq C_6^{-1} \left(-\inf_M f - \varepsilon/2 \right)$$

so $-\inf_M f$ is bounded in terms of $\|f\|_{L^1}$. $\square$

<u>Step 3</u> Bounding $\partial\bar{\partial} f$ (not quite $\|f\|_{C^2}$)

Let $\tilde{g}_{j\bar{k}} = g_{j\bar{k}} + \partial_j \partial_{\bar{k}} f$, so $\det(\tilde{g}_{j\bar{k}}) = e^F \det(g_{j\bar{k}})$

$\mathrm{tr}_g \tilde{g} = g^{j\bar{k}} \tilde{g}_{j\bar{k}}$, $\mathrm{tr}_{\tilde{g}} g = \tilde{g}^{j\bar{k}} g_{j\bar{k}}$

$\lambda_1, \dots, \lambda_n$ = eigenvalues of $\tilde{g}$ w.r.t. g
(of endomorphism $A^i{}_j = g^{i\bar{k}} \tilde{g}_{j\bar{k}}$)

$\mathrm{tr}_g \tilde{g} = \lambda_1 + \dots + \lambda_n$, $\mathrm{tr}_{\tilde{g}} g = \frac{1}{\lambda_1} + \dots + \frac{1}{\lambda_n}$, eq^n: $\lambda_1 \cdots \lambda_n = e^F$

<u>Key inequality</u> $\exists B > 0$ (dep. on (M, g))

such that

$$(*) \qquad \tilde{\Delta} \log \operatorname{tr}_g \tilde{g} \geq -B \operatorname{tr}_{\tilde{g}} g - \frac{g^{j\bar{k}} \tilde{R}_{j\bar{k}}}{\operatorname{tr}_g \tilde{g}}$$

$\tilde{\Delta} u = \tilde{g}^{j\bar{k}} \partial_j \partial_{\bar{k}} u$, $\quad \tilde{R}_{j\bar{k}}$ = Ricci curv. of $\tilde{g}$.

<u>Bounding $\operatorname{tr}_g \tilde{g}$ assuming $(*)$</u>:

- $\tilde{\Delta} f = \tilde{g}^{j\bar{k}} (\tilde{g}_{j\bar{k}} - g_{j\bar{k}}) = n - \operatorname{tr}_{\tilde{g}} g$
- Cauchy-Schwarz ineq:

$$(\operatorname{tr}_g \tilde{g})(\operatorname{tr}_{\tilde{g}} g) \geq n^2$$

$$\text{so} \quad (\operatorname{tr}_g \tilde{g})^{-1} \leq \frac{\operatorname{tr}_{\tilde{g}} g}{n^2}$$

- $\tilde{R}_{j\bar{k}} = -\partial_j \partial_{\bar{k}} F + R_{j\bar{k}}$

$$\overset{(*)}{\Longrightarrow} \quad \tilde{\Delta} \log \operatorname{tr}_g \tilde{g} \geq -B \operatorname{tr}_{\tilde{g}} g + \frac{\overbrace{\boxed{g^{j\bar{k}}(\partial_j \partial_{\bar{k}} F - R_{j\bar{k}})}}^{\text{bounded}}}{\operatorname{tr}_g \tilde{g}}$$

$$\geq -C \operatorname{tr}_{\tilde{g}} g$$

$$\tilde{\Delta}\Big(\underbrace{\log \operatorname{tr}_g \tilde{g} - (C+1) f}_{=h}\Big) \geq \operatorname{tr}_{\tilde{g}} g - (C+1) n$$

Let $h(p) = \max_M h$. Then $\tilde{\Delta} h(p) \leq 0$

so $\operatorname{tr}_{\tilde{g}} g(p) \leq (C+1) n$

$$\Rightarrow \quad \mathrm{tr}_g \tilde{g}(p) \leq C_1$$

$$\Rightarrow \quad h(p) \leq C_2 \quad (\text{dep on } \|f\|_{C^0})$$

$$\Rightarrow \quad \max_M h \leq C_2$$

$$\Rightarrow \quad \mathrm{tr}_g \tilde{g} \leq C_3 \quad \text{on } M$$

$\square$